

Mathematics, 2010 EC, entrance exam

Grade 12 · Mathematics · 2010 E.C.

Time allowed: 120 minutes

Questions: 57

Total marks: 57

Name _____ Grade _____ ID No. _____ Date _____

1. What is the equation of the line that passes through (1, 1) and is parallel to the line $3y - x = 1$? (1)
- A. $3x - y = 2$ B. $x + 3y = 4$
C. $3y - x + 2 = 0$ D. $x - 3y + 2 = 0$
-
2. Which one of the following is an equation of the circle whose end points of a diameter are (0, -2) and (2, 2)? (1)
- A. $x^2 + y^2 - 2x - 4 = 0$ B. $x^2 + y^2 = 4$
C. $(x - 1)^2 + y^2 = 4$ D. $x^2 + y^2 - 2y - 4 = 0$
-
3. What is the maximum value of the function $f(x) = x^4 - 2x^2$ on $[-2, 1]$? (1)
- A. 8 B. 12
C. 24 D. 40
-
4. If the truth value of $(p \wedge \neg p) \Leftrightarrow [(q \vee \neg q) \Rightarrow r]$ is True, then which one of the following must be True? (1)
- A. p B. q
C. $\neg q$ D. $\neg r$
-
5. Let A and B be two events. Suppose that the probability that neither event occurs is $3/8$. What is the probability that at least one of the event occur? (1)
- A. $5/8$ B. $1/4$
C. $1/8$ D. $3/4$
-
6. What is the value of $\sum_{n=2}^{20} \left(\frac{1}{n-1} - \frac{1}{n} \right)$ (1)
- A. $\frac{17}{20}$ B. $\frac{21}{20}$
C. $\frac{19}{20}$ D. $\frac{23}{20}$
-
7. Which one of the following is a convergent sequence? (1)
- A. $\left\{ \frac{(-1)^n}{2} \right\}$ B. $\left\{ \frac{1}{n} + \sin(n) \right\}$
C. $\left\{ \frac{1-3^n}{2^n} \right\}$ D. $\left\{ \frac{1+2^n}{3^n} \right\}$
-
8. If a function f is differentiable at a , then what is the value of $\lim_{x \rightarrow a} f(x)$ (1)
- A. $f'(a)$ B. $f(a)$
C. $f'(a) - f(a)$ D. 0
-
9. If $f'(x) = e^{x-1} + 3x^2 - \frac{1}{x}$ and $f(1)=5$, what is $f(x)$? (1)
- A. $f(x) = e^{(x-1)} + x^3 - \ln x + 3$ B. $f(x) = e^{(x-1)} - x^3 + \ln x + 5$
C. $f(x) = e^{x-1} + 3x^2 - \frac{1}{x} + 5$ D. $f(x) = e^{x-1} + 3x^2 - \frac{1}{x^2} + 2$

10. If A is a square matrix of order 3 and $\det(A)=5$, then what is the value of $\det(A.\text{adj}(A))$? (1)

A. 3
B. 5
C. 25
D. 125

11. A salesman sold items x_1, x_2 and x_3 , with different rates of commissions as shown in the table below. What are the rates of commission on items x_1, x_2 and x_3 , respectively. (1)

[Figure: A table with a header row reading: Months | Sales of Unit (split into three sub-columns X_1, X_2, X_3) | Total Commission (in Birr). Data rows: February, $X_1 = 90, X_2 = 100, X_3 = 20$, total commission 800; March, $X_1 = 130, X_2 = 50, X_3 = 40$, total commission 900; April, $X_1 = 60, X_2 = 100, X_3 = 30$, total commission 850.]

A. 4, 2 and 11
B. 4, 11 and 2
C. 2, 4 and 11
D. 11, 2 and 4

12. If $z = -3 + 4i$ and $w = 1 + 2i$, then what is the value of $\frac{2z}{w} + \overline{w}$? (1)

A. $2 + 3i$
B. $3 + 2i$
C. $3 + 5i$
D. $3 - 2i$

13. The population of a certain city is increasing at a rate of 3% per year. If the population was 100,000 in 2010 E.C., then what will be the population in 2020 E.C? (Given: $(1.03)^9 = 1.30, (1.03)^{10} = 1.34, (1.3)^9 = 10.60, (1.3)^{10} = 13.78$) (1)

A. 130,000
B. 134,000
C. 1,060,000
D. 1,378,000

14. What is the slope of the tangent line to the graph of $f(x) = 3e^x + \sin x + 2$ at the point (0, 5)? (1)

A. 2
B. 3
C. 4
D. 5

15. What is the value of $\int_1^2 \frac{x+4}{x(x+2)} dx$ (1)

A. 2
B. $\ln 2$
C. $\ln 3$
D. $\ln 4 - \ln 3$

16. What is the area of the region enclosed by the graph of $y^2 = x + 1$ and $y^2 = -x + 1$? (1)

A. $8/3$ sq.units
B. $4/3$ sq. units
C. $3/8$ sq. units
D. $3/4$ sq. units

17. What is the partial fraction decomposition of $\frac{x^2+x+1}{(x+2)(x^2+1)}$? (1)

A. $\frac{5}{3(x+2)} + \frac{2x+1}{3(x^2+1)}$
B. $\frac{3}{5(x+2)} + \frac{2x+1}{5(x^2+1)}$
C. $\frac{2}{5(x+2)} + \frac{3x+1}{5(x^2+1)}$
D. $\frac{2}{3(x+2)} + \frac{2x+1}{3(x^2+1)}$

18. What is the area of the triangle (in sq. units) formed by the lines joining the vertex of the parabola $x^2 = -36y$ to the end points of the latus rectum? (1)

A. 126
B. 162
C. 216
D. 261

19. The time needed to type a sample of 8 business letters in an office is 7, 8, 6, 8, 9, 7, 5, 6 minutes. What are the mean ($\bar{x}$) and the standard deviation(s) of the data in minute? (1)

A. $\bar{x} = 7, s = \sqrt{2}$
B. $\bar{x} = 7, s = \sqrt{1.5}$
C. $\bar{x} = 8, s = \sqrt{2}$
D. $\bar{x} = 8, s = \sqrt{1.5}$

20. Let A be a 3×3 invertible matrix and B any be any 3×3 matrix. If $|A| = a$ and $|B| = b$, then which one of the following is NOT true? (1)
- A. $|A^T A| = a^2$ B. If $b = 0$, then B is not invertible.
 C. $|KA| = K^3 |A|$, for any $K \in \mathbb{R}$ D. $|A^{-1}B| = ab$
-
21. Let $z = \left(\frac{1-i}{1+i}\right)^{18}$. Then what is the value of z ? (1)
- A. i B. -1
 C. $-i$ D. $1 - i$
-
22. If $f(x) = \frac{|x|}{x}$ and $g(x) = \frac{x+2}{x^3-4x}$, then what is the value of $\lim_{x \rightarrow -2} (f(x) + g(x))$? (1)
- A. $-9/8$ B. $-7/8$
 C. $9/8$ D. ∞
-
23. what is the value of $\lim_{x \rightarrow 0} \frac{1}{x^2} \sin^2\left(\frac{x}{2}\right)$? (1)
- A. $1/4$ B. $1/2$
 C. 2 D. 4
-
24. Which one of the following is NOT true about the function $f(x) = 3x^4 - 4x^3$? (1)
- A. $(0, 0)$ is point of inflection of f .
 B. 0 and 1 are critical numbers of f .
 C. f is decreasing on $(-\infty, 1)$ and increasing on $(1, \infty)$.
 D. f is concave upward on $(0, 2/3)$ and concave downward on $(-\infty, 0)$ and on $(2/3, \infty)$
-
25. What is the value of $\int x\sqrt{1-x^2} dx$? (1)
- A. $\frac{1}{2}\sqrt{1-x^2} + c$ B. $-\frac{1}{3}(1-x^2)^{3/2} + c$
 C. $(1-x^2)^{3/2} + c$ D. $-2x\sqrt{1-x^2} + c$
-
26. What is the value of $\int_1^9 \frac{e^{\sqrt{x}}}{\sqrt{x}} dx$? (1)
- A. $e^3 - e$ B. $\frac{e^3}{3} - e$
 C. $2(e^3 - e)$ D. $-2x\sqrt{1-x^2} + c$
-
27. A man running a race-course noted that the sum of the distance from the two flag posts from him is always 10 meters and the distance between the flag post is 8 meters. What is the equation of the path traced by the man? (1)
- A. $\frac{x^2}{81} + \frac{y^2}{25} = 1$ B. $\frac{x^2}{64} + \frac{y^2}{100} = 1$
 C. $\frac{x^2}{100} + \frac{y^2}{64} = 1$ D. $\frac{x^2}{25} + \frac{y^2}{9} = 1$
-
28. Suppose the following are the premises of an argument. He is healthy and he is not angry. He is angry or he is plan fails. His plan does not fail if he does not travel abroad. Given that the premises are true, which one of the following can be a conclusion that makes the argument valid? (1)
- A. He travels abroad.
 B. His plan fails and he is angry.
 C. His plan does not fail.
 D. His plans fails and he does not travel abroad.

29. The age distribution of students in a certain class is given below: What is the modal value of the distribution? (1)
 [Figure: A two-row frequency table. Row 1, labelled Age, gives the class intervals 10 - 14, 15 - 19, 20 - 24 and 25 - 29. Row 2, labelled No. of Students, gives the matching frequencies 2, 10, 6 and 7.]
- A. 17.38
 B. 17.83
 C. 18.37
 D. 18.73
-
30. Let $A = \{1, 2, 3, 4, 5, 6, 7\}$, $B = \{7, 8, 9\}$ and $C = \{8, 9, 10\}$. If one of the number is deleted randomly from each of these sets, what is the probability that all the three deleted numbers are even or are multiples of 3? (1)
- A. $1/9$
 B. $2/21$
 C. $8/63$
 D. $4/5$
-
31. Let A be a 3×3 matrix and $|A| = -2$. Then what is the value of $|\text{adj}(A)|$? (1)
- A. 4
 B. -2
 C. $-1/2$
 D. -8
-
32. If the sum of the first three consecutive terms of an arithmetic progression $\{A_n\}$, with $A_n > 0$ for all n , is 9 and the sum of their squares is 35, then what is the sum s_n of the first n terms? (1)
- A. $n^2 + 1$
 B. $n^2 - 1$
 C. $2n^2 + 1$
 D. n^2
-
33. What is the value of $\int \frac{1}{x} (\ln x + x^2 e^{-x}) dx$? (1)
- A. $\frac{1}{2} \ln^2 x - (x + 1)e^{-x} + c$
 B. $\frac{1}{2} \ln^2 x + (2 - x)e^{-x} + c$
 C. $\frac{1}{2} x^2 \ln x + (2 - x)e^{-x} + c$
 D. $\frac{1}{2} x^2 \ln x - (x + 1)e^{-x} + c$
-
34. Let $f(x) = \frac{3x+1}{x-2}$. Then what is the range of $f(x)$? (1)
- A. $\mathbb{R} \setminus \{2\}$
 B. $\mathbb{R}$
 C. $\mathbb{R} \setminus \{-1/3\}$
 D. $\mathbb{R} \setminus \{3\}$
-
35. Which one of the following is true about the graph of $f(x) = \frac{2x^3+2x^2+3x}{x^2+x}$? (1)
- A. The graph has y-intercept at (0, 3).
 B. The vertical asymptote of the graph is only $x = -1$ and its oblique asymptote is $y = 2x$.
 C. The graph has at least one x-intercept.
 D. The vertical asymptotes of the graph are at $x = 0$ and $x = -1$ but it has no horizontal asymptote.
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36. If $z = (1 + \sqrt{3}i)(1 + i)$, then which one of the following is the polar representation of z ? (1)
- A. $z = 2\sqrt{2} (\cos(105^\circ) + i \sin(105^\circ))$
 B. $z = 2\sqrt{2} (\cos(15^\circ) + i \sin(15^\circ))$
 C. $z = 4 (\cos(105^\circ) + i \sin(105^\circ))$
 D. $z = 4 (\cos(75^\circ) + i \sin(75^\circ))$
-
37. What is the sum of the series $\sum_{n=1}^{\infty} 3^n 4^{-n}$? (1)
- A. ∞
 B. $3/6$
 C. 4
 D. 3
-
38. Let $f(x) = \ln(x\sqrt{x})$. Then what is $f'(x)$ equal to? (1)
- A. $2x/3$
 B. $\sqrt{x}/2$
 C. $3/2x$
 D. $2/x\sqrt{x}$

39. What is the maximum possible area of a rectangle in square units with diagonal of length 16 units? (1)
- A. 128
B. 64
C. 48
D. 256
-
40. Let $U = \mathbb{N}$ (the set of natural numbers) be the universe. Which one of the following proposition is true? (1)
- A. $(\exists x)(x + x = x)$
B. $(\forall x)(\exists y)(x - y = x)$
C. $(\forall x)(\exists y)(y < x)$
D. $(\forall x)(\exists y)(x \div y = y \div x)$
-
41. What are the values of a and b so that the function $f(x) = \begin{cases} x + 1, & x < 1 \\ ax + b, & 1 \leq x < 2 \\ 3x, & x \geq 2 \end{cases}$ is continuous everywhere? (1)
- A. $a = 4, b = 2$
B. $a = 4, b = -2$
C. $a = -4, b = -2$
D. $a = -4, b = 2$
-
42. A cylindrical tank whose inner diameter is 2m contains $4\pi \text{ m}^3$ oil. If the oil discharged from the tank at the rate of $(2\pi/3) \text{ m}^3/\text{min}$, then how long (in min) does it take for the tank to be empty? (1)
- A. $4/3$
B. 4
C. 6
D. 12
-
43. Let $f(x) = x - x^2$ and $g(x) = 1/x$. Then what is $g(f(\frac{1}{x}))$ equals to (1)
- A. $\frac{x^2}{x-1}$
B. $x - x^2$
C. $\frac{1}{x^2 - x}$
D. $\frac{x-1}{x^2}$
-
44. The variance of 20 observation is 5. If each observation is multiplied by 2, then what is the variance of the resulting observations? (1)
- A. 5
B. 10
C. 40
D. 20
-
45. If $f(x) = k \ln x + e^{\sin x}$ and $f(\pi) = -1$, then what is the value of k? (1)
- A. π
B. $2\pi^2$
C. π^2
D. 2π
-
46. If θ is the fourth quadrant angel and $\sec \theta = \sqrt{2}$ then what is $\csc \theta$ equals to? (1)
- A. $\frac{1}{\sqrt{2}}$
B. $\frac{-1}{\sqrt{2}}$
C. $-\sqrt{2}$
D. $\sqrt{2}$
-
47. Consider the following assertion: $P + 2^n$ is an odd number for any prime p and any $n \in \mathbb{N}$. Which one of the following is correct about a prove or disprove of the assertion? (1)
- A. There is a counter example that disproves the assertion.
B. The assertion can be proved by direct method; because p is odd and $2^n = 2(2^{n-1})$ is even imply that $p + 2^n$ is odd since the sum of odd and even is odd.
C. The assertion can be proved by the indirect method: because if $n \notin \mathbb{N}$, then $2^n \notin \mathbb{N}$ and hence $p + 2^n$ is not odd
D. The assertion can be disproved by the method of contradiction.

48. What is the standard equation of the line passing through the point (2,3) and parallel to the line given by (1)

$$\begin{cases} x = 1 + 2\lambda \\ y = -2 - \lambda \end{cases}, \lambda \in \mathbb{R}?$$

- A. $\frac{x-2}{-1} = \frac{y-3}{2}$ B. $\frac{x-1}{1} = \frac{y-3}{-2}$
C. $\frac{x-1}{2} = \frac{y-3}{-2}$ D. $\frac{x-2}{2} = \frac{y-3}{-1}$

49. Let $P(n)$ be an open proposition on the set of natural numbers (N). Which one of the following is a correct application of the principle of mathematical induction? (1)

- A. If $P(1)$ is true for $n = 1$; and if both $P(n)$ and $P(n+1)$ are true for certain $n \in N$, then $P(n)$ is true for all $n \in N$.
B. If $P(10)$ is true; and if $p(n)$ is true implies that $p(n+1)$ is true, then $p(n)$ is true for all $n \in N$.
C. If $p(10)$ is true; and assuming $p(n)$ is true for any $n > 10$ it follows that $p(n+1)$ is true, then $p(n)$ is true for all $n \geq 10$
D. If $p(1)$ is true; and $p(n) \Rightarrow p(n+1)$ is true for any $n \in N$, then $p(n)$ is true for all $n \in N$.

50. If the image of the line $2x - 3y = 7$ under a translation is $2x - 3y = 0$, which one of the following is a translation vector of the translation line? (1)

- A. $U = (-2, 1)$ B. $U = (1, -2)$
C. $U = (-1, 2)$ D. $U = (2, -1)$

51. Let $P = (1, \alpha, \alpha)$ and $Q = (\alpha - 1, 1, 1)$ be two points in space and the distance between P and Q is 3. Then what is the value(s) of α ? (1)

- A. $\alpha = -3, \alpha = 1$? B. $\alpha = 1, \alpha = -9$
C. $\alpha = -1, \alpha = 9$ D. $\alpha = 3, \alpha = -1/3$

52. What is the image of the circle $x^2 + y^2 - 4x - 6y + 12 = 0$ when it is reflected with respect to the line $y = -x$? (1)

- A. $(x - 2)^2 + (y - 3)^2 = 1$ B. $(x + 3)^2 + (y + 2)^2 = 1$
C. $(x + 2)^2 + (y + 3)^2 = 1$ D. $(x - 3)^2 + (y - 2)^2 = 1$

53. If the dot products of a vector $\vec{A}$ with the vectors $i - j + k$, $2i + j - 3k$ and $i + j + k$ are 4, 0 and 2, respectively, what is $\vec{A}$? (1)

- A. $\vec{A} = (2, 1, 1)$ B. $\vec{A} = (-2, 1, -1)$
C. $\vec{A} = (2, -1, 1)$ D. $\vec{A} = (-2, -1, 1)$

54. In order to measure the height of a tower, suppose a surveyor takes two sightings from a transit 1 meter high which are positioned d meter s apart on the same ground level as in the figure below. If the first measured angle of elevation is α and the second is β and d ? (1)

[Figure: A side-on survey diagram. A tall tower stands upright at the right of the picture on level ground. Two transit stands sit on the same ground level to the left of it, each drawn as a short vertical post: the farther one, at the far left, is labelled 2 and the nearer one is labelled 1. A dashed line of sight runs from the top of each post up to the top of the tower. The angle of elevation at the farther post is marked beta and the angle at the nearer post is marked alpha. A horizontal dimension line with end ticks spans the ground between the two posts and is labelled d .]

- A. $\left(\frac{\tan \alpha \tan \beta}{\tan \alpha - \tan \beta} \right) d + 1$ B. $\frac{\tan \alpha \tan \beta}{(\tan \alpha - \tan \beta)} + 1$
C. $\left(\frac{\tan \alpha - \tan \beta}{\tan \alpha \tan \beta} \right) d + 1$ D. $\left(\frac{\tan \alpha \tan \beta}{\tan \alpha + \tan \beta} \right) d + 1$

55. Let l be the line given by the vector equation $(x, y) = (-2, 1) + \lambda(1, 1), \lambda \in \mathbb{R}$, which one of the following is the equation of the image of l after being translated by the vector $u=(2, -1)$ followed by a rotation through 45° about the origin? (1)

A. $y = -2\sqrt{2}x$

B. $y = \sqrt{2}$

C. $y = \sqrt{2}x$

D. $x = 0$

56. what is the period (p) and the range of $f(x) = 5 \sin\left(\frac{1}{3}x + 2\right) + 3$? (1)

A. $P = 6\pi$, range = $[-5, 5]$

B. $P = 6\pi$, range = $[-2, 8]$

C. $p = \frac{2\pi}{3}$, range = $[-5, 5]$

D. $p = \frac{3}{2}\pi$, range = $[-2, 8]$

57. What is the value of $\lim_{n \rightarrow \infty} \sum_{k=1}^n \frac{2}{n} \left[\left(\frac{2k}{n}\right)^3 + 5 \left(\frac{2k}{n}\right) \right]$? (1)

A. 14

B. 10

C. 4

D. 18

— END OF EXAMINATION —

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